TT–I–545a
FEBRUARY 29, 1956
SUPERSEDING
Int. Fed. Spec. TT–I–00545a (GPO)
Revision 1, November 5, 1955, and
Fed. Spec. TT–I–545
November 28, 1947

FEDERAL SPECIFICATION

INK; MARKING, QUICK-DRYING, FOR PHOTOGRAPHIC FILM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — The quick-drying marking ink covered by this specification is intended only for marking film. It is not a drawing ink.

1.2 Classification.

1.2.1 *Type and grade.*—The ink shall be of one type and grade.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications. — Federal Specification TT–P–141, Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing, of the issue in effect on date of invitation for bids, forms a part of this specification.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Standards. — Military Standard MIL–STD–129, Marking for Shipment and Storage, of the issue in effect on date of invitation for bids, forms a part of this specification.

(Copies of Military standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. — The manufacturer is given wide range in the selection of materials and in the process of manufacture, provided that the ink produced meets the requirements of this specification.

3.2 Properties.—The ink shall be a special waterproof ink that will not spread or blot and will adhere readily when used for marking the uncoated sides of films which have a cellulose acetate base. The ink shall flow readily under all conditions of humidity when used in a steel writing pen or in lettering devices and numbering machines. Coagulation or precipitation shall not take place in the ink under ordinary working conditions nor when the ink is stored for 6 months at temperatures in the range of 5° to 40°C.

3.3 Color.—The color of the ink shall be black.

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3.4 Density. — Impressions of the ink shall be clearly legible through negatives which have densities of 1.0 or less.

3.5 Drying time.—At 25° ± 5°C. and 65 ± 2 percent relative humidity, the ink shall dry in not more than 30 seconds when it is applied to film with a new Esterbrook No. 048 steel pen or one of equivalent fineness.

3.6 Solubility.—After the ink has dried completely, it shall be insoluble in cold water or in commonly used photographic fixing solution.

3.7 Solvent. — A suitable solvent shall be provided by the manufacturer for making erasures and for cleaning pens. The solvent shall not adversely affect the film base or emulsion, in any manner, when used to remove inked images from the film. The amount of solvent to be furnished shall be as specified by the procuring agency.

3.8 Permanence and durability. — Markings shall not fade, become brittle, nor become detached from the film when tested as specified in 4.4.4.

3.9 Flash point. — The flash point shall be not less than 200° F. (open cup).

3.10 Toxicity.—The ink and its solvent shall be non-injurious to workers exposed to atmospheric concentrations as great as 500 ppm, for an 8-hour working day.

3.11 Etching. — The ink shall not etch the film when applied.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Two samples, each not less than 1 ounce, shall be taken at random from each delivery.

4.2 Inspection. — Inspection shall be made by the Government to determine compliance with the requirements of this specification.

4.3 Material for making tests.—Tests shall be made on photographic film which has been properly developed, fixed, washed, and dried. All tests shall be made on the uncoated side of the film.

4.4 Tests.

4.4.1 *Density.* — A film negative having a density of 1.0 shall be lettered with a new Esterbrook No. 048 steel pen or one of equivalent fineness. A photographic print shall be made from this negative. The letters on the resulting print shall be inspected for legibility.

4.4.2 *Lettering tests.* — The lettering tests shall be made with a clean Esterbrook No. 048 steel writing pen or one of equivalent fineness on cellulose acetate film.

4.4.2.1 Dip the pen in the ink and draw on the uncoated film five parallel lines each 6 inches long. The lines shall have uniform density and width throughout their length. The ends shall show no increase in width.

4.4.2.2 Draw on the uncoated film a line 6 inches long. One hour after the ink has dried, immerse the film for 1 hour in water having a temperature of not more than 21°C. The ink shall not become soluble, and the line shall remain legible.

4.4.3 *Erasing solvent.*—Rub a dried mark on the film with a clean cloth which has been moistened with the solvent. The ink shall be removed easily and completely without etching the film and shall not leave streaks, milkiness or visible residue.

4.4.4 *Permanence and durability.* — Mark a film with lettering and cut the film in two parts so that each part has about one-half of the lettering. Expose one part to the radiation from a Fade-Ometer, Type FDA-R, or its equivalent, for 24 hours. Afterward, compare the lettering on the exposed film with that on the portion of the film which was not exposed. There shall be no appreciable fading of the lettering. The markings shall not be brittle nor be detached from the film.

4.3.5 *Flash point.* — The flash point of the ink shall be determined as described in method 429.4 of Federal Specification TT-P-141.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. — Unless otherwise specified, the ink and the solvent shall be furnished in 1-ounce circular glass bottles which shall be closed with a suitable screwcap. Each bottle shall be individually packaged.

5.2 Packing.—Unless otherwise specified, 48 packaged bottles shall be packed in a corrugated fiberboard container.

5.2.1 *Domestic shipment.* — When domestic shipment is specified, interior packages shall be packed in suitable, uniform commercial exterior shipping containers so constructed as to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation. Containers shall be constructed of corrugated or solid fiberboard having a minimum Mullen test of 200 pounds, conforming to the requirements of Consolidated Freight Classification Rules in effect at time of shipment. Containers shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

5.3 Marking.

5.3.1 *Packages.* — Unless otherwise specified, each bottle and each individual container shall be marked with the following data, as applicable:

Ink; Marking, Quick-Drying, for Photographic Film
Federal Specification number
Color
Quantity
Stock number
Order number
Manufacturer's name or trade-mark
Information regarding a suitable solvent, and marked: "CAUTION—DO NOT FREEZE"

Solvent; for Photographic Film Marking Ink
Federal Specification number
Quantity
Stock number
Order number
Manufacturer's name or trade-mark

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked as follows:

5.3.2.1 *Civil agencies.*—Each container shall be marked with the name of the commodity, the quantity contained therein as defined by the contract or order under which the shipment is made, the name of the contractor, and the contract or order number. In addition, each container shall be marked "CAUTION—DO NOT FREEZE".

5.3.2.2 *Military.* — Each container shall be marked in conformance with Military Standard MIL–STD–129, and in addition shall be marked "CAUTION—DO NOT FREEZE".

6. NOTES

6.1 Ordering data.—Purchasers should specify the amount of solvent required and should exercise any desired options offered herein. (See 3.7, 5.1, 5.2, and 5.3.)

6.2 Transportation description.—Transportation description applicable to this item is:

Ink, printing, not otherwise indexed by name
Carload minimum weight 36,000 pounds.
Truckload minimum weight 36,000 pounds.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal Specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

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Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—Q M
Navy—A S
Air Force.

U. S. GOVERNMENT PRINTING OFFICE: 1956

Copies of this specification may be purchased for 10 cents each.

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